

## STERLING FORGE CONSTRUCTION CO.

2900 Wesleyan St, Suite 400, Houston, TX 77027 | Lic: CSLB-102341 | DBE/HUB Certified

### BID - LUXURY MIXED-USE TOWER FOUNDATION & PODIUM PACKAGE - VALUE ENGINEERED OPTION

**Date:** September 20, 2026

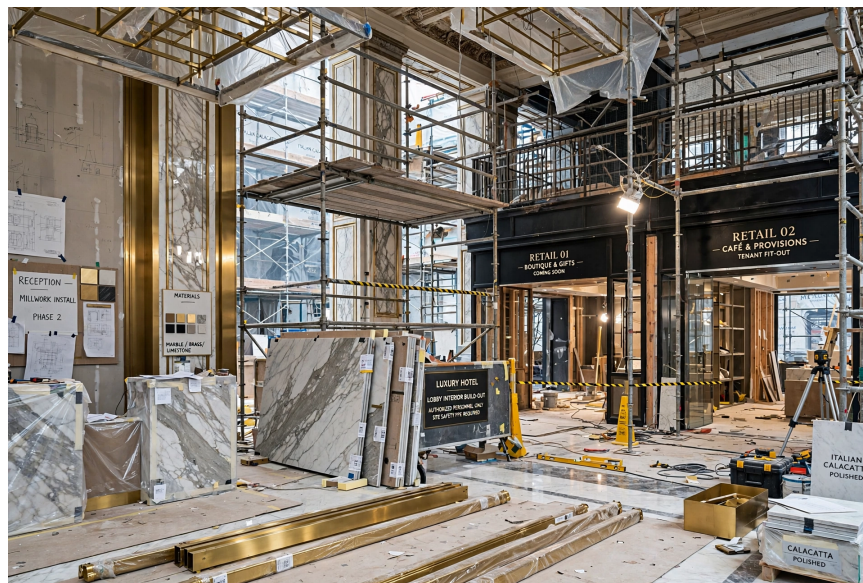
**Project:** Lumina Hotel & Residences – Foundation & Podium (Same scope as Atlas/Meridian)

**Owner:** BlueRock Institutional Partners LP

**Proposal ID:** SFC-2026-FND-067C | Valid 90 Days Firm

**Bid Amount:** \$87,200 (Base VE) – Lowest Cost, Precast-Integrated Approach

**Philosophy:** Value engineering for institutional investors focused on cost discipline without compromising safety – use precast grade beam shells + site-cast infill, reduced PT tonnage, alternative waterproofing.



### PAGE 1: EXECUTIVE SUMMARY & VALUE ENGINEERING

Sterling Forge Construction is a Houston-based DBE contractor with 18 years in mid/high-rise foundations (completed 32 podium slabs). For Lumina, we propose a value-engineered foundation/podium at \$87,200 – \$11,550 below Atlas and \$25,200 below Meridian – by leveraging precast grade beam stay-in-place forms (Gate Precast), reduced PT from 33 to 28 tendons via higher concrete strength, and economical waterproofing. This is same scope intent but alternative means: drilled piers remain 36-inch straight shaft (no bell) to 48 ft to avoid belling tool rental and risk in caving soils (per geotech, straight shaft to deeper depth acceptable with 1800 psf skin friction + end bearing). Podium: 11-inch (vs 12-inch) with 6000 psi + 7000 psi at column strips. Waterproofing: Cetco Voltex DS bentonite + HDPE. We maintain ACI 318 compliance and include geotech review for straight shaft. This VE saves rental of belling tool (\$3,500) and shortens pier drilling by eliminating bell cleaning.

### PAGE 2: ENGINEERING, MATERIALS & PRECAST INTEGRATION

**Concrete:** Pier: 4500 psi @28d (vs 5000/5500), Type I/II, 20% fly ash, w/c 0.45, 6-8 inch slump (non-SCC, saves \$45/CY). Podium: 6000 psi (vs 6000/6500), 0.42 w/c, fiber only microfiber (no macrofiber saves \$800). Strength at stressing: 75% f'c ~4500 psi.

**Rebar:** Grade 60 throughout, #6 @ 12-inch podium, #7 verticals (10 bars per pier vs 12/14), #4 ties @ 18-inch (vs 12-inch) per ACI min, epoxy only at grade beam top. Rebar shop drawings included but not 3D modeled (2D CAD).

**PT:** 0.5-inch 270K unbonded, 28 tendons @ 12-ft spacing vs 32-36, reduced tonnage 1.0 ton, encapsulated. Single-stage stressing at 100% f'c (simpler).

**Precast Grade Beams:** 30x36 precast U-shells (4-inch wall) set on piers, then infill with 4000 psi concrete + rebar. Saves

formwork labor \$2,800 and speeds schedule 2 days. Connection: dowels from pier cage into shell, grout.

**Waterproofing:** Voltex DS bentonite geotextile (self-healing) + 6mm protection + dimple board (standard flow). No redundant subdrain – single 4-inch perf PVC. Sump: single Zoeller (no duplex). Savings \$3,000 vs Carlisle.

**Geotech:** PSI report allows straight shaft alternative to 48 ft depth (deeper than Atlas 40 ft). Includes 1 site visit. No extra borings.

### PAGE 3: MEANS & METHODS & EQUIPMENT

**Equipment:** Texoma 650 drill rig (smaller than Bauer/Soilmec), 36-inch auger straight shaft, 25-ton crane (vs 50-ton), 32m boom pump (vs 36/42m). Lower mobilization cost.

**Sequence:** 32 days (4 days longer than Atlas, 11 longer than Meridian). Week 1: Mob, excavation, dewatering (2 wells, passive, no telemetry, 40 gpm). Week 2-3: Drill 12 piers straight shaft 48 ft (slower due to deeper depth, 1.5 per day), no belling, set precast shells. Week 3-4: Grade beam infill, podium formwork (Symons steel-ply, not Peri aluminum), PT install (28 tendons), pour 260 CY (11-inch vs 12-inch = 8% less concrete = \$1,200 savings). Week 5: Stressing, backfill, demob.

**Logistics:** Laydown 30x50 ft (smaller), rebar from Harris Supply (local), concrete from Vulcan Materials (4-mile). No drone, no QR codes.

**Safety:** EMR 0.82, OSHA 10, excavation competent person, PT barricades. No third-party safety audit, daily checklist.

### PAGE 4: QAI/QC, COMPLIANCE & DOCUMENTATION

**Testing:** Terracon: slump, air, temp per 2 trucks (vs per truck), 4 cylinders per 100 CY (vs 50 CY), PIT on 50% piers (6 of 12) vs 100%, no CSL, no GPR, no maturity sensors. Elongation records per tendon.

**Compliance:** 2021 IBC, ACI 318-19. No LEED EPDs, no structural peer review included (optional \$2,200). ACI 301. Geotech straight shaft alternative requires SER approval – allowance \$800 for SER review (vs \$1,500-2,500 others).

**BIM:** 2D CAD only, no BIM, no 4D, no clash detection beyond visual. One coordination meeting included. Procore daily reports but no P6, no Earned Value.

**Environmental:** SWPPP basic, dewatering discharge to storm with city permit, no turbidity monitoring included, silt fence.

### PAGE 5: SCHEDULE, WARRANTY & RISK

**Schedule:** 32 days NTP to stressing, 7 weather days included (vs 5/10). No liquidated damages proposed; instead 5% retention until 60 days. No early bonus.

**Warranty:** 1-year workmanship (Texas minimum), 1-year concrete, 10-year PT tendon (supplier), 2-year waterproofing (Voltex) vs 5-year Carlisle. No extended warranty option.

**Insurance:** GL \$1M/\$2M, Umbrella \$2M, Auto \$1M, Workers Comp statutory. No pollution, no builders risk, no professional liability. OCIP compatible but no Pollution coverage for dewatering.

**Risk & Exclusions:** Excludes obstruction allowance (if hit old foundation, \$175/hr extra), excludes soil export beyond 100 CY (vs 200 CY), excludes permanent dewatering, elevator pit, structural steel, fireproofing, lightning, mass excavation. Straight shaft depth increase to 48 ft adds risk if bearing stratum deeper than 48 ft – requires change order at \$95/LF extra if deeper. No allowance for unforeseen obstructions – time & material.

### PAGE 6: PRICING - VALUE ENGINEERED

| Item                  | VE Scope                                         | Cost     |
|-----------------------|--------------------------------------------------|----------|
| Piers 12ea (straight) | 36-inch straight 48ft, 10 bars, 4500psi, PIT 50% | \$31,500 |
| Grade Beams Precast   | U-shell precast + infill 4000psi                 | \$10,200 |
| Podium PT Slab        | 11-inch 6200 sqft, 6000psi, 28 tendons           | \$26,800 |
| Waterproofing VE      | Voltex bentonite + protection + single drain     | \$5,200  |
| Dewatering Basic      | 2 wells passive, 3 weeks                         | \$2,100  |
| Engineering           | SER allowance \$800, 2D CAD                      | \$900    |
| Mob                   | Texoma rig + 25T crane + 32m pump                | \$3,100  |
| SUBTOTAL              | Base                                             | \$79,800 |
| Contingency           | For deeper pier if needed (est)                  | \$2,400  |
| Overhead & Profit     | 12% (vs 15-18% others)                           | \$10,000 |
| TOTAL                 | Net ex-tax firm 90 days                          | \$87,200 |
| Option                | Upgrade to 100% PIT + 2 CSL                      | \$2,800  |
| Option                | Duplex sump + BMS                                | \$1,900  |

**Payment:** Net 25, 25% mob, 50% pier, 25% podium. Retainage 10% until 60 days (vs 30 days). Bond at 3.0% extra (higher due to smaller balance sheet). No early pay discount, no early award credit.

**Price Lock:** 90 days firm (best among bids), no escalation clause – good for investor budgeting.

**Why Cheapest:** Savings from: straight shaft no bell (\$3,500), precast shells (\$2,800), 11-inch slab (8% less concrete \$1,200), fewer tendons (\$1,800), cheaper waterproofing (\$3,000), reduced testing (PIT 50% vs 100%, no CSL/GPR \$2,500), 2D vs BIM (\$1,500), lower insurance limits, smaller equipment. Total VE savings ~\$18k vs Atlas.